

Experiment 4: Fixed Height + Obstacle + Multi Wind Strength | weak_wind | Comparison Table

Dropped-Only Quality

Method	N(drop)	Precision Mean(m)	Precision P50(m)	Precision P90(m)	R@1m	R@2m	R@3m	Roll Mean	Roll P90	Pitch Mean	Pitch P90	Impact Mean	Impact P90	Impact P95
PPO	2000	0.972	0.712	1.768	67.6%	93.3%	98.5%	7.796	12.271	3.473	6.588	0.294	0.451	0.547
PPO-GRU	2000	2.464	0.497	12.873	81.8%	86.8%	87.1%	2.654	6.102	3.081	5.395	0.242	0.415	0.473
PPO-RNN	1999	0.768	0.686	1.148	82.4%	98.7%	98.8%	5.858	12.180	3.467	6.847	0.234	0.344	0.398
MPC	1655	2.438	2.545	2.903	0.0%	18.2%	99.5%	1.503	2.485	1.210	2.255	0.133	0.135	0.139

No-Drop Breakdown (done but no release)

Method	N(done)	N(drop)	Drop Rate	N(no_drop)	No-drop Rate	Crash Rate	Crash+NoDrop	Crash in NoDrop	Timeout+NoDrop	Both Flags
PPO	2000	2000	100.0%	0	0.0%	0.0%	0	nan	0	0
PPO-GRU	2000	2000	100.0%	0	0.0%	0.0%	0	nan	0	0
PPO-RNN	2000	1999	100.0%	1	0.1%	0.1%	1	100.0%	0	0
MPC	2000	1655	82.8%	345	17.2%	17.2%	345	100.0%	0	0

Cross-Case Check: PPO/PPO-GRU/PPO-RNN on MPC No-Drop Cases (large error > 3.0m)

Method	In MPC-NoDrop	Dropped	Drop Rate	Err>3.0m	LargeErr Rate (in dropped)	Mean Err (in dropped)
PPO	345	345	100.0%	5	1.4%	0.928
PPO-GRU	345	345	100.0%	33	9.6%	2.038
PPO-RNN	345	345	100.0%	2	0.6%	0.713